PROJECT PROPOSAL

University Archive & Management System

A Framework-Based Web Application for Academic Project Archiving and Management

Submitted for

Software Development Project – II (SDP-II)

Submitted By

Nilay Dav Nath

Department of Computer Science and Engineering

Submitted To

Department of Computer Science and Engineering

2026

1. Introduction

The University Archive & Management System is a web-based academic management platform designed to digitize, organize, preserve, and manage university academic projects and related information in a centralized environment.

In many academic institutions, project documents and academic records are maintained using traditional paper-based processes, personal storage devices, or scattered digital files. Such approaches can make it difficult to locate previous projects, monitor project submissions, manage approvals, and preserve academic resources for future use.

The proposed system provides a centralized platform where students can submit their academic projects, professors can review and manage submitted projects, and administrators can monitor users, projects, departments, activities, and system statistics.

The system will be developed as a modern framework-based web application using the Laravel framework with PHP. The frontend will use HTML5, CSS3, JavaScript, and Bootstrap to provide a responsive and user-friendly interface. MySQL will be used for database management.

2. Background

Academic projects are important resources for universities because they represent the practical knowledge, research activities, and technical achievements of students. However, without a centralized digital archive, previously completed projects can become difficult to locate and manage.

The University Archive & Management System aims to provide a structured digital repository for academic projects. Students will be able to submit project information and files, while professors will be able to review and approve submissions. Administrators will have control over users, departments, projects, and system activities.

The system will also provide searching and filtering facilities so that archived projects can be found using relevant information such as department, year, keywords, supervisor, and project title.

3. Problem Statement

Traditional academic project management methods can create several difficulties for students, professors, and university administrators. These problems include:

- Difficulty in storing academic projects in a centralized location.
- Loss or duplication of project documents.

- Difficulty in finding previously completed projects.
- Manual project submission and approval processes.
- Lack of centralized project status tracking.
- Difficulty in monitoring student project activities.
- Limited access to academic project history.
- Lack of organized project metadata.
- Difficulty in generating academic and administrative reports.

The proposed system addresses these issues by providing a centralized web-based platform for academic project submission, review, approval, searching, archiving, and management.

4. Motivation

The main motivation behind this project is to improve the way university academic projects are stored and managed.

A centralized digital archive can make academic resources easier to access and preserve. Students can use previous projects as academic references, professors can efficiently monitor project submissions, and administrators can manage project-related information from a single platform.

Another motivation is to develop the system using a modern PHP framework. Laravel provides an organized development structure based on the Model-View-Controller (MVC) architecture, which can improve maintainability, security, scalability, and code organization.

5. Aim of the Project

The main aim of the University Archive & Management System is to develop a centralized, secure, and user-friendly web application for managing, archiving, reviewing, searching, and preserving university academic projects.

The system will provide separate functionalities for students, professors, and administrators while maintaining controlled access to academic information.

6. Objectives

The major objectives of the proposed system are:

1. To develop a centralized digital archive for university academic projects.
2. To allow students to upload and manage their academic projects.
3. To allow professors to review, approve, reject, or request modifications to submitted projects.
4. To provide advanced project searching and filtering facilities.
5. To maintain structured project metadata.
6. To provide role-based access for students, professors, and administrators.
7. To provide dashboards containing useful academic and system statistics.
8. To provide notifications related to project submission and approval activities.
9. To generate academic and administrative reports.
10. To develop the system using the Laravel PHP framework.
11. To create a responsive user interface using Bootstrap.
12. To design a maintainable and scalable system using MVC architecture.

7. Scope of the Project

The proposed system will focus on academic project archiving and management within a university environment.

The major areas covered by the system are:

7.1 Student Management

Students will be able to:

- Register and log into the system.
- Manage their profile information.
- Upload academic projects.
- Provide project metadata.
- View submitted projects.
- Track project approval status.
- Receive project-related notifications.
- Search archived projects.

7.2 Professor Management

Professors will be able to:

- Log into the professor portal.
- View assigned student projects.
- Review submitted projects.
- Approve projects.
- Reject projects.
- Request modifications.
- Monitor project status.
- Communicate with students regarding projects.

7.3 Administration

Administrators will be able to:

- Manage students and professors.
- Manage departments and academic information.
- Monitor project submissions.
- Monitor system activities.
- View dashboard statistics.
- Generate reports.
- Manage archived projects.

8. Proposed System Features

The major features of the system are described below.

8.1 User Authentication

The system will provide secure authentication for different types of users. Users will log in using valid credentials and will only be able to access the features allowed for their roles.

8.2 Project Upload

Students will be able to submit academic projects along with relevant metadata such as:

- Project title
- Abstract
- Department
- Subject
- Supervisor
- Academic year
- Project file

8.3 Project Approval

Professors will review submitted projects. A project may be approved, rejected, or returned to the student for modification.

8.4 Project Search and Filtering

Users will be able to search archived projects using different criteria, including:

- Project title
- Keywords
- Department
- Academic year
- Supervisor
- Subject

8.5 Dashboard and Analytics

The system will provide dashboards for different users. Dashboard information may include:

- Total users
- Total projects

- Approved projects
- Pending projects
- Rejected projects
- Department-wise statistics

8.6 Notification System

The system will provide notifications for important project-related activities such as:

- Project submission
- Project approval
- Project rejection
- Modification request
- Important system updates

8.7 Report Generation

Administrators will be able to generate useful academic and activity reports based on available system data.

9. User Roles

The system will mainly contain three categories of users.

Table 1: User Roles and Responsibilities

User	Major Responsibilities
Student	Upload projects, manage profile, view project status, search archived projects, and receive notifications.
Professor	Review submitted projects, approve or reject projects, request modifications, and monitor assigned students.
Administrator	Manage users, departments, projects, system activities, reports, and overall system configuration.

10. Technology Stack

The proposed system will use the following technologies:

Table 2: Technology Stack

Technology	Purpose
PHP	Server-side programming language
Laravel	PHP web application framework
HTML5	Web page structure
CSS3	Styling and layout
JavaScript	Client-side interaction and dynamic functionality
Bootstrap	Responsive and modern user interface design
MySQL	Relational database management
Apache / XAMPP	Local development server environment
Blade	Laravel template engine
Git / GitHub	Version control and source code management

11. Framework and Architecture

The system will be developed using the Laravel PHP framework. Laravel provides a structured development environment and supports the Model-View-Controller (MVC) architectural pattern.

The major components will be:

- **Model:** Handles database-related operations and application data.
- **View:** Provides the user interface using Blade templates, HTML, CSS, JavaScript, and Bootstrap.
- **Controller:** Handles application logic and connects user requests with models and views.
- **Routes:** Define the URLs and requests handled by the application.
- **Middleware:** Controls access to protected resources and role-based functionality.

The proposed architecture can be represented as:

User Interface

HTML5 + CSS3 + JavaScript + Bootstrap

↓

Blade Templates

↓

Laravel Application
Routes + Controllers + Middleware + Models

↓

MySQL Database

12. Database Overview

The system will use MySQL as the relational database management system.
The major entities are expected to include:

- Users
- Students
- Professors
- Departments
- Projects
- Subjects
- Project Approvals
- Notifications
- Messages

The database will maintain relationships between users, projects, departments, subjects, supervisors, and project approval records.

13. Functional Requirements

The system will provide the following major functional requirements:

1. The system shall allow authorized users to log in.
2. The system shall provide role-based access.

3. Students shall be able to submit projects.
4. Students shall be able to view project submission status.
5. Professors shall be able to review submitted projects.
6. Professors shall be able to approve or reject projects.
7. Professors shall be able to request modifications.
8. Users shall be able to search archived projects.
9. Administrators shall be able to manage users.
10. Administrators shall be able to manage academic departments.
11. Administrators shall be able to monitor project activities.
12. The system shall provide notifications for important events.
13. The system shall provide dashboard statistics.
14. The system shall support report generation.

14. Non-Functional Requirements

14.1 Security

The system should provide secure authentication, authorization, password protection, input validation, and secure file upload mechanisms.

14.2 Performance

The system should provide reasonable response time and should support multiple users accessing the application.

14.3 Usability

The user interface should be simple, responsive, and easy to navigate.

14.4 Maintainability

The Laravel MVC architecture will help maintain the application through modular and organized code.

14.5 Scalability

The system should be designed so that additional departments, users, projects, and future modules can be incorporated.

14.6 Reliability

The system should maintain data consistency and provide suitable error handling and recovery mechanisms.

15. Development Methodology

The project will follow an iterative development approach. The development process will be divided into several stages:

1. Requirement Analysis
2. System Design
3. Database Design
4. UI Design
5. Laravel Application Development
6. Module Integration
7. Testing
8. Debugging
9. Deployment
10. Documentation

An iterative approach will allow the system to be developed module by module and tested continuously throughout the development process.

16. System Models

The following system models will be developed during the project:

- Use Case Diagram
- Entity Relationship Diagram (ERD)

- Data Flow Diagram (DFD)
- Activity Diagram
- Sequence Diagram
- Class Diagram
- System Architecture Diagram

These models will help describe the functional behavior, data relationships, workflows, and architectural structure of the proposed system.

17. Testing Plan

Testing will be performed to ensure that the developed system works correctly and satisfies the defined requirements.

The following testing approaches will be considered:

- Unit Testing
- Integration Testing
- System Testing
- Functional Testing
- Security Testing
- User Acceptance Testing

Each major module will be tested individually before integration with the complete system.

18. Feasibility Analysis

18.1 Technical Feasibility

The system is technically feasible because it will use widely available technologies such as PHP, Laravel, MySQL, HTML, CSS, JavaScript, and Bootstrap.

18.2 Economic Feasibility

The system can be developed using open-source technologies and commonly available development tools, reducing software licensing costs.

18.3 Operational Feasibility

The system is designed for students, professors, and administrators. Its web-based interface will allow users to access the required functions through a modern web browser.

18.4 Schedule Feasibility

The project can be developed incrementally by dividing the work into requirement analysis, design, development, testing, and documentation phases.

19. Expected Outcome

After successful implementation, the University Archive & Management System is expected to provide:

- A centralized digital academic project archive.
- Faster project searching and retrieval.
- Organized project submission and approval workflow.
- Better communication between students and professors.
- Role-based management of academic information.
- Improved monitoring of project activities.
- Responsive and user-friendly interfaces.
- Better maintainability through Laravel MVC architecture.
- Improved academic resource preservation.

20. Project Timeline

The tentative development schedule is presented below.

Table 3: Tentative Project Schedule

Task	**Estimated Duration**
Requirement Analysis	Week 1–2
System and Database Design	Week 2–3
UI Design with Bootstrap	Week 3–4
Laravel Project Setup	Week 4
Authentication and User Management	Week 5
Student Module	Week 6–7
Professor Module	Week 7–8
Admin Module	Week 8–9
Project Archive and Search	Week 9–10
Notification and Reporting	Week 10–11
Testing and Debugging	Week 11–12
Final Documentation	Week 12–13

21. Future Enhancements

The system may be extended in the future with the following features:

- Mobile application.
- AI-assisted academic project search.
- Intelligent project recommendation.
- Email and push notification integration.
- Cloud-based deployment.
- Advanced academic analytics.
- Integration with university academic systems.
- Advanced document similarity checking.

These features may be considered after the successful implementation of the core system.

22. Conclusion

The University Archive & Management System is proposed as a centralized web-based platform for managing and preserving university academic projects. The system will provide students, professors, and administrators with dedicated functionalities for project submission, review, approval, searching, archiving, monitoring, and reporting.

The use of the Laravel framework will provide a structured MVC-based development environment, while Bootstrap will support the development of a responsive and user-friendly interface. PHP will be used as the server-side programming language and MySQL will be used for database management.

The proposed system is expected to reduce the difficulties associated with traditional academic project management and improve the accessibility, organization, and preservation of academic resources.

23. References

1. Laravel Documentation. *Laravel – The PHP Framework for Web Artisans*.

2. PHP Documentation. *PHP: Hypertext Preprocessor*.

3. MySQL Documentation. *MySQL Reference Manual*.

4. Bootstrap Documentation. *Bootstrap – Front-end Framework*.

5. Mozilla Developer Network. *HTML, CSS and JavaScript Documentation*.

6. IEEE. *Software Requirements Specification Documentation Guidelines*.